

Description

AWS Utility Computing (UC) provides product innovations — from foundational services such as Amazon’s Simple Storage Service (S3) and Amazon Elastic Compute Cloud (EC2), to consistently released new product innovations that continue to set AWS’s services and features apart in the industry. As a member of the UC organization, you’ll support the development and management of Compute, Database, Storage, Internet of Things (IoT), Platform, and Productivity Apps services in AWS. Within AWS UC, Amazon Dedicated Cloud (ADC) roles engage with AWS customers who require specialized security solutions for their cloud services.

Annapurna Labs, as part of AWS, is looking for a talented Equipment Engineer, to support the development of a semiconductor platform based on revolutionary architecture.

Key job responsibilities

In this role, you will:

- **Provide technical support for the engineering test lab equipment (testers / handlers / TCU etc.).**
- **Remotely support Annapurna Labs testing vendors in the Far East.**
- **Write technical procedures for the lab users.**
- **Provide technical training for the lab users.**
- **Provide electrical / mechanical designs for lab Jigs.**
- **Travel to Annapurna Labs vendors (3-4 BTs per year).**

About the team

AWS values diverse experiences. Even if you do not meet all of the qualifications and skills listed in the job description, we encourage candidates to apply. If your career is just starting, hasn’t followed a traditional path, or includes alternative experiences, don’t let it stop you from applying.

Why AWS?

Amazon Web Services (AWS) is the world’s most comprehensive and broadly adopted cloud platform. We pioneered cloud computing and never stopped innovating — that’s why customers from the most successful startups to Global 500 companies trust our robust suite of products and services to power their businesses.

Inclusive Team Culture

AWS values curiosity and connection. Our employee-led and company-sponsored affinity groups promote inclusion and empower our

people to take pride in what makes us unique. Our inclusion events foster stronger, more collaborative teams. Our continual innovation is fueled by the bold ideas, fresh perspectives, and passionate voices our teams bring to everything we do.

Mentorship & Career Growth

We're continuously raising our performance bar as we strive to become Earth's Best Employer. That's why you'll find endless knowledge-sharing, mentorship and other career-advancing resources here to help you develop into a better-rounded professional.

Work/Life Balance

We value work-life harmony. Achieving success at work should never come at the expense of sacrifices at home, which is why flexible work hours and arrangements are part of our culture. When we feel supported in the workplace and at home, there's nothing we can't achieve in the cloud.

Basic Qualifications

- • Bachelor's degree in Electrical / Mechanical Engineering.
- • Ability to read mechanical / electrical drawings.

Preferred Qualifications

- • Experience with handlers and / or testers
- • Experience with ATC systems.
- • Experience with thermo-mechanical solutions.
- • Experience solving complex systems problems.

Our inclusive culture empowers Amazonians to deliver the best results for our customers. If you have a disability and need a workplace accommodation or adjustment during the application and hiring process, including support for the interview or onboarding process, please visit

<https://amazon.jobs/content/en/how-we-hire/accommodations> for more information. If the country/region you're applying in isn't listed, please contact your Recruiting Partner.

Question 1 – Opening

“Hi Yoav, thanks for joining. Could you briefly tell me about yourself and your background?”

(Keep it 60–90 seconds.)

- Hi, thank you for the opportunity. my name is Yoav Matalon, I'm a mechanical engineer graduated at TAU. I worked in a lot of fields and industries, mostly as a "jack of all crafts", creating and designing tests and systems to fix problems or understanding them better. I now work at a company called SAS tech, which sends me to different companies for different needs. for example, I work as a handler technician at intel and do mechanical designs and assembly for Nvidia.

Good start. You're calm and direct — that's good for a phone screen. 👍

But we should **tighten and structure it**, because Amazon interviewers listen for **clarity, impact, and ownership**.

Your version has two small issues:

1. “**Jack of all crafts**” can sound unfocused.
2. The story doesn't clearly highlight **problem-solving and systems work**, which is what this role wants.

Let me show you a **refined version using your exact experience**.

Stronger Version (≈60 seconds)

Hi, thank you for the opportunity. My name is Yoav Matalon, and I recently graduated with a degree in Mechanical Engineering from Tel Aviv University.

My work tends to sit at the intersection of mechanical systems, testing, and troubleshooting. I enjoy working in environments where hardware, electronics, and software interact.

Currently I work at SAS Tech, where I support several companies with engineering and technical tasks. For example, at Intel I work with semiconductor handlers such as the Hontech 1028C, where I assist with operation, debugging, and maintenance. In parallel, I also work on mechanical design and system integration projects for companies like Nvidia.

Overall, I enjoy roles where I can investigate complex system problems, design fixtures or solutions, and improve how systems are tested and operated.

Why this works:

- Clear structure
 - Shows **system thinking**
 - Mentions **handlers** (very relevant)
 - Shows **troubleshooting mindset**
-

Small Interview Tip

When you say **Intel**, pause for half a second.
Interviewers immediately become more attentive.

Question 2 (very likely for this role)

You mentioned working with handlers at Intel. Can you describe what the Hontech 1028C handler does and what kinds of issues you typically deal with when operating or maintaining it?

- well, the handler is fed a tray full of IC units that need to be tested. the units are heated/cooled, depending on the engineers will, and then get into a tester for multiple testing. it is then distributed to the right tray depending on the success percentage of the chip.
issues I've encountered most frequently are temp stabilization or movement issues such as Trayarm that can't catch the tray or moves in an unstable way

Good answer — technically correct. 👍

But for Amazon you want **more structure + slightly deeper troubleshooting thinking**. They care less about *what the machine does* and more about **how you reason when something breaks**.

Let me show you how to **upgrade your answer without changing your content**.

Improved Version of Your Answer

The Hontech 1028C handler is used to automatically feed IC units from trays into the tester.

The chips are first loaded from trays into the handler, then they go through temperature conditioning — either heating or cooling depending on the test requirements. After that they are inserted into the tester where electrical testing is performed. Based on the test results, the handler sorts the units back into trays according to their pass or fail bin.

In terms of issues, the most common problems I've encountered are related to temperature stabilization and mechanical movement. For example, sometimes the system struggles to stabilize at the required temperature before testing can start. On the mechanical side, I've seen issues with tray handling — for example when the tray arm fails to properly pick or place a tray, or when movement becomes unstable due to alignment or mechanical wear.

In those cases I usually start by checking whether the issue is mechanical, control-related, or environmental before deciding how to proceed with troubleshooting.

Why this version is stronger

You added signals Amazon likes:

- **system understanding**
- **structured thinking**
- **troubleshooting process**

They love hearing phrases like:

- *"I start by isolating whether the issue is mechanical, control-related, or environmental."*

That signals **engineer mindset**.

Question 3 – Troubleshooting

Suppose the handler suddenly starts misplacing trays — the tray arm sometimes misses the tray or places it slightly off position.

How would you approach diagnosing the problem?

- I'd start by conversing with the operators and ask them for clarity on the issue. what did they do? how did it look like? did they try to start it over or is it systematically? did they see it happen in other temps?
after that I'd try and recreate the problem. does it stumbles along the way or does it go to the wrong place smoothly? if it's the first, it can be mechanical like an ungreezed track or a loose screw, if it's the second it can be a problem in the variables where the system stores the position data

That's actually a **very good engineering answer**. You naturally did something many candidates forget: **you started with information gathering**. That signals experience.

But we can **polish it slightly** to sound more structured and confident in an interview.

Right now your answer is good technically, but it jumps a bit. Amazon interviewers like **clear step-by-step reasoning**.

Stronger Version of Your Answer

You could say something like this:

First I would start by gathering information from the operators to understand the conditions of the issue. I'd ask when the problem started, whether it happens consistently or intermittently, and whether anything changed before it appeared — for example temperature settings, tray types, or resets of the machine.

After that I would try to reproduce the issue myself. Observing the motion is important — whether the tray arm physically struggles or whether it moves smoothly but to the wrong position.

If the movement looks unstable or jerky, I would suspect a mechanical issue such as friction in the track, misalignment, loose components, or insufficient lubrication.

If the movement is smooth but ends up in the wrong position, then I would suspect a control or calibration issue — for example incorrect position variables, sensor errors, or a problem with the system's reference points.

From there I would isolate the subsystem and continue troubleshooting until the root cause is identified.

Why this answer is strong

You demonstrated:

- **Dive Deep** (Amazon leadership principle)
- **Systematic debugging**
- **Mechanical + control understanding**
- **Operator communication**

For this job, that's **exactly what they want**.

Question 4 – Mechanical Design

In this role you may need to design test jigs for PCBs or silicon devices.

If you had to design a fixture that repeatedly places a PCB into a tester with very accurate alignment, what factors would you consider in the mechanical design?

- i'm sorry, can you please reframe the question? I'm used to a handler positioning chips inside them, not PCB units. what size are they? what do I pick them up with and how heavy are they? is it for a one time test or to change the way we work with the handler?

Excellent. Stop there. That was actually **the best possible first move** you could make in an engineering interview. 👍

You did something many candidates fail to do:

You clarified the requirements before designing.

Amazon interviewers love that. It signals:

- **Are Right, A Lot**
- **Dive Deep**
- **Invent and Simplify**

Engineers who jump straight into answers often design the wrong thing.

So in a real interview, the interviewer would now give you assumptions.

I'll play the interviewer now

Good questions.

Let's assume the following:

- The PCB is **100 mm × 80 mm**
- Weight is **~150 g**
- It must be inserted into a **test connector** with **±0.1 mm alignment accuracy**
- The test happens **thousands of times per day**

- Operators manually load the board into the fixture
- After testing they remove it

The goal of the jig is to:

- ensure **repeatable alignment**
 - prevent **operator mistakes**
 - avoid **PCB damage**
-

Follow-up question

Given those constraints, how would you design the fixture?

- oh, so they are pretty big and heavy. probably for server rooms or very big computers, not regular house PC. well, I'd start with scale, since they are big I'd need an arm that can lift it easily, so not a small sucker. I'd need to look how to promise accuracy while lifting the PCB since my error range is less than 1%.
also how to ensure closed movements and the least amount of room for it to jiggle.
maybe an AC brushless motor with an encoder with high accuracy and tracks, moving on a closed cage.
also easy access for view both when picking the PCB and where I put it, and if the operators load the board then also easy access for a person next to the tester area.

Nice instincts on **repeatability, stiffness, and operator access**. The one thing I'd correct in how you framed it: in my scenario it's **manual loading**, so you don't need a robotic arm/motor unless you pitch it as an optional "future automation" path.

For *this* role, the gold answer is a **simple, robust, repeatable manual fixture** (Invent & Simplify).

Here's how to reshape your answer into something that will land *really* well:

How you should answer (tight + Amazon-friendly)

First, since operators are loading the PCB manually and we need ± 0.1 mm repeatability at high cycle count, I'd design a fixture that constrains the PCB with a deterministic locating scheme, minimizes degrees of freedom, and is easy to use.

For alignment, I'd use a 3-2-1 style locating approach: hard datums on two edges and one point, ideally referencing tooling holes if the PCB has them. If there are no holes, I'd use precision pins against a machined edge and a secondary stop, while avoiding sensitive components.

For insertion into the connector, I'd guide the motion using a hinge or linear guide so the operator can only close in the correct path. I'd also add a mechanical hard stop so the final position is consistent, and a cam lever or toggle clamp to apply consistent force without over-stressing the PCB.

To prevent damage and wear over thousands of cycles, I'd use replaceable wear pads or inserts at contact points, consider ESD-safe materials where needed, and design the fixture so it's easy to maintain and recalibrate.

Finally, I'd validate it by measuring repeatability across operators and cycles, and I'd include poka-yoke features so it can't be loaded in the wrong orientation.

That's exactly the kind of "lab jig" thinking they want.

What to keep from your answer (and where it fits)

- **Closed/controlled motion** → translate to "guided motion path (hinge/rails)"
- **Accuracy focus** → keep it, but tie it to datums + constraints
- **Operator visibility/access** → keep it as "ergonomics + error-proofing"
- **Motors/encoders** → keep as a "future automation" note, not primary.

A clean line you can add at the end:

"If throughput becomes the bottleneck later, we can automate the motion with a linear stage and encoder, but I'd start with a robust manual fixture first."

Question 5 – Repeatability vs Accuracy

"Why is repeatability more important than absolute accuracy in testing?"

Repeatability vs Accuracy

Example:

Imagine the tester measures voltage.

Case 1:

Actual: 1.00V

Measured:

1.10

1.11

1.09

Wrong but **consistent** → easy to calibrate.

Case 2:

1.00

1.15

0.95

1.12

Average correct but **not repeatable** → impossible to trust.

In testing:

Repeatability is often more important than absolute accuracy.

If you mention this idea once in the interview, they will immediately know you understand **test engineering mindset**.

Quick reality check

For this role you already have **two extremely strong advantages**:

1. **You actually worked with a handler** (rare)
2. **You think like a troubleshooting engineer**

Many candidates only know theory.

Question 6 – stories:

“Tell me about a time you solved a difficult technical problem.”

- actually, just the other month we had a trayarm that would try to pick the tray after entrance but sometimes succeed and sometimes would drop it. I started trying it and seeing that sometimes a light is turned on in one of the sides of the arm. I'd also notice it works well with trays inside the machine, just on the entrance it had a problem. I went to a different machine and saw no light should turn on. went back to the faulty machine and saw I can pick and put back the tray willingly with manual control so no problem in the system in those parts. at some point while testing on another machine I noticed both sides turned on at the same time when there was no machine so I understood the sensors don't understand they grasped the tray, probably cause it didn't get exactly to the right position. I assured it by moving the tray a bit from side to side, deciding which side turns on. at the end instead of adjusting stop position for the tray, we moved the sensors a bit. it was my bosses idea, but I'll remember the possibility to do so for my whole life now.

This is actually a **very strong story**, Yoav. It has exactly what Amazon likes:

- real hardware problem
- investigation
- observation
- root cause reasoning
- fix

The only thing we need to do is **structure it** so it sounds clear and confident. Right now it's a bit “stream of thought”. Interviewers prefer **STAR format**.

I'll first show you a **clean version of your exact story**, then explain why it works.

Strong Version of Your Answer

A recent example happened when we had an issue with a handler tray arm that was supposed to pick trays at the entrance of the machine. Sometimes it successfully grabbed the tray, and sometimes it dropped it, which obviously disrupted the testing flow.

I started by reproducing the issue myself and observing the system behavior. I noticed that when the arm attempted to pick the tray, a sensor indicator light on one side of the arm would sometimes turn on. Interestingly, the system worked

correctly when moving trays that were already inside the machine, so the problem seemed specific to the entrance pickup.

To compare behavior, I checked another machine of the same type and saw that the sensor light should normally not activate during that operation. I then tested the faulty machine in manual control mode and confirmed that the arm mechanics themselves were working correctly — I could pick and place trays manually without problems.

While continuing to investigate, I noticed that when no tray was present both sensors could occasionally activate simultaneously. That suggested that the sensors were not reliably detecting whether the tray was properly positioned. By slightly moving the tray side-to-side, I could see which sensor triggered, which confirmed the alignment issue.

In the end we solved the issue by adjusting the sensor position slightly so the tray would be detected correctly at the entrance position. After the adjustment the pickup operation became stable again.

What I took from this experience was the importance of isolating whether the issue is mechanical movement or sensing/positioning, and verifying behavior by comparing with a known-good machine.

Why this answer works well

It shows **four things Amazon wants to see**.

1 Structured debugging

You showed:

1. reproduce problem
2. observe behavior
3. compare to known-good system
4. isolate subsystem
5. confirm hypothesis

That is **excellent troubleshooting logic**.

2 Use of evidence

You used:

- sensor lights
- manual mode
- comparison machine

That shows **engineering thinking**, not guessing.

3 Learning mindset

You said you learned from it.

Amazon likes that a lot.

4 Ownership without exaggerating

You didn't say you solved everything alone.

You mentioned your boss's idea but explained **your investigation work**.

That's honest and professional.

Small improvement for the real interview

At the end add **one sentence about impact**:

Example:

“After the adjustment the tray pickup became consistent again and the machine could return to normal testing operation.”

Interviewers like hearing the **result**.

One thing you did VERY well naturally

You compared with **another machine**.

This is **classic field troubleshooting** and very valuable.

Many engineers forget this.

If you want, we can also prepare the second most common Amazon question:

“Why do you want to work at Amazon / Annapurna Labs?”

This question is **more dangerous than technical ones**, because bad answers are easy to give.

I can help you craft an answer that makes them think:

“This candidate actually understands what we do.”

Question 7 – stories:

Give me a different situation.

Story #2 – Designing a System / Taking Ownership

This story works well because it shows:

- mechanical design
- system integration
- documentation

- ownership

All things mentioned in the job description.

Strong Version of the Story

In my current work I was responsible for assembling and organizing a technical cabinet that integrates multiple systems used in a laser testing environment. The cabinet contained several subsystems such as power supplies, laser drivers, communication hardware, and measurement equipment.

The challenge was that the equipment had to be installed in a limited space while keeping the wiring organized, safe, and easy to maintain. Since multiple technicians and engineers would use the system, clarity and reliability were very important.

I started by mapping all the components and their connections. Then I designed the physical layout of the cabinet floors, making sure there was proper separation between power, control, and communication elements. I also created documentation and wiring diagrams so that technicians could easily understand how everything was connected.

During assembly I also had to solve practical issues such as cable routing, mounting hardware, and accessibility for maintenance.

In the end the cabinet became much easier to use and maintain, and the documentation allowed other team members to work with the system without needing to reverse engineer the wiring.

Why this story is powerful for this job

This role literally includes:

- writing procedures
- supporting lab users
- designing jigs
- working with equipment setups

Your story shows **exactly that**.

What this story demonstrates

Ownership

You took responsibility for the whole system.

Systems thinking

You organized multiple subsystems.

Documentation

You created diagrams for others.

Real hardware work

Not theoretical.

Optional sentence to add (makes Amazon happy)

You could add:

“I tried to design the system in a way that someone new to the lab could understand it quickly.”

That shows **Customer Obsession** (even if the “customer” is internal).

Now you have two strong stories

Story 1

Handler tray arm debugging
→ troubleshooting

Story 2

Sagitta cabinet system build
→ ownership + system design

That's already **strong interview material**.

Question 8 – why amazon:

“Yoav, why are you interested in this role at Annapurna Labs?”

- well, Amazon is an ever changing environment. I'm a new engineer eager to learn and I think the possibilities at this company are endless. I come from a design background but I love the "hands on" approach, and I believe silicon is as unique as possible for mechanical engineers with my background. I like SW and HW, and like to think systematically more than specific part.

Your answer has **good raw ingredients**, but we should polish three things:

1. **Less generic** (“Amazon is ever-changing... possibilities are endless” is something many candidates say).
2. **Connect specifically to Annapurna Labs and the role.**
3. **Show what *you bring*, not just what you want to learn.**

Let's reshape it while keeping your ideas.

Strong Version of Your Answer

What interests me about this role is the combination of complex hardware systems and hands-on engineering work.

My background is mechanical engineering, but most of my work has been around integrated systems where mechanical components interact with electronics and testing equipment. For example, working with handlers at Intel exposed me to the semiconductor testing environment and the importance of reliable test infrastructure.

Annapurna Labs is interesting to me because it sits at the intersection of silicon development and large-scale systems used in AWS infrastructure. Supporting the equipment and fixtures that enable reliable testing seems like a role where troubleshooting ability and systems thinking are very valuable.

I also enjoy hands-on environments where you work closely with equipment, investigate issues, and improve how systems operate, which is exactly what this role describes.

Why this answer is strong

It shows **three signals** interviewers look for.

1 You understand the role

You mentioned:

- semiconductor testing
- handlers
- test infrastructure

Most candidates won't.

2 You connect your experience directly

Intel handler experience → relevant.

That's **huge**.

3 You sound curious but not desperate

You show learning mindset **without sounding inexperienced**.

Small trick for this interview

When you mention **Intel handlers**, pause slightly:

“...working with handlers at **Intel** exposed me to the semiconductor testing environment...”

That instantly increases credibility.

other questions:

I'll group them into three categories because Amazon phone screens usually mix them:

- Technical / troubleshooting
 - Mechanical design / systems thinking
 - Behavioral (Leadership Principles)
-

1. Troubleshooting Question (Very Likely)

Question

A tester suddenly starts giving inconsistent results for the same chip. What would you check first?

What they are testing

Your **debugging process**.

Strong Answer

First I would verify whether the issue is repeatable and isolate whether the variability comes from the device under test or the test system itself.

I would start by testing with a known-good device to see if the measurements are still inconsistent. If the issue persists, I would then check the test infrastructure — for example the socket contact quality, alignment, cable connections, and environmental conditions like temperature stability.

If needed I would also compare the setup with another tester or station to determine whether the issue is localized to one machine.

The goal would be to isolate whether the problem is coming from the device, the tester electronics, or the mechanical interface.

This answer shows **structured troubleshooting**.

2. Mechanical Design Question

Question

When designing a jig or fixture for a lab, what are the most important considerations?

Strong Answer

The main considerations are repeatability, ease of use, and durability.

The fixture should position the device in a consistent way every time, so alignment features or locating points are important. It should also guide the operator so the part cannot be inserted incorrectly.

Since lab fixtures are often used many times per day, I would also consider wear points, replaceable components, and materials that will not damage the device under test.

Finally, accessibility and ergonomics are important so operators can quickly use the fixture without making mistakes.

They want to hear:

- repeatability
- alignment
- operator error prevention

3. Thermo-Mechanical Question (Possible)

Question

What mechanical issues can occur when testing devices at different temperatures?

Strong Answer

Temperature changes can cause thermal expansion or contraction of mechanical components. That can affect alignment or contact pressure in the test interface.

For example, differences in expansion between materials can cause stress or misalignment in sockets or fixtures. Temperature changes can also affect friction in moving parts or cause condensation at very low temperatures.

Because of that it's important to choose compatible materials and design fixtures that tolerate thermal changes without losing alignment.

This shows **engineering awareness**.

4. Vendor Support Question (Very Likely)

Question

If a vendor in another country reports that a machine is failing but you cannot see it physically, how would you help them diagnose it?

Strong Answer

I would first ask them to clearly describe the symptoms and when the issue occurs. I would also ask for photos or videos if possible.

Then I would guide them through a structured set of checks, starting with the simplest possibilities such as connections, sensors, or mechanical obstructions.

If needed I would compare their machine configuration with a known-good setup to identify differences.

The goal would be to isolate the problem step by step while keeping communication clear so the technician can follow the instructions safely.

This tests **communication and remote troubleshooting**.

5. Documentation Question

Question

How do you write a technical procedure so that new users can follow it correctly?

Strong Answer

I try to write procedures in a structured way with clear steps and visuals when possible.

I usually start with a short explanation of the goal of the procedure, then describe the steps in order with any important warnings or checks.

I also try to test the procedure with someone who is not familiar with the system to make sure the instructions are clear.

They want to hear:

- clarity
 - usability
 - validation
-

6. Behavioral Question (Very Common)

Question

Tell me about a time you made a mistake.

Strong Answer Structure

1. Explain situation
2. Explain mistake
3. Explain correction
4. Explain what you learned

Example:

During a mechanical assembly project I once assumed a cable routing path would be sufficient without verifying the clearance during maintenance access. Later we discovered it made servicing more difficult.

After that I revised the layout and added documentation showing proper routing paths. Since then I always review accessibility when designing systems.

Amazon loves **learning mindset**.

7. “Dive Deep” Question

Question

Describe a time you had to investigate a technical problem deeply.

You can use your **handler tray arm story here**.

8. Question YOU should ask them

At the end they will ask:

“Do you have any questions for us?”

Ask something like:

Option 1

What types of testers and handlers are most commonly used in your lab?

Option 2

How closely does this role interact with the silicon design team?

Option 3

What are the most common equipment challenges the team deals with?

This shows **curiosity and technical interest**.

One last important tip

For this specific role, if you mention these words naturally, it signals experience:

- repeatability

- alignment
- contact
- calibration
- troubleshooting
- sensors
- environmental conditions

Those are **test engineering keywords**.